Unit 5.1 – Basic Care and Comfort: Key Concepts from Lewis (Ch. 10, 19, 20, 22)

Chapter 10 – Pain

- **Pain definition and magnitude**: Pain is an unpleasant sensory and emotional experience associated with actual or potential tissue damage; it is subjective and best assessed by the patient's self-report.[^1]
- Pain is multidimensional (physiological, sensory-discriminative, motivational-affective, behavioural, cognitive-evaluative, sociocultural), and effective management requires attention to all dimensions.[^1]
- Unrelieved pain is common in Canada and has major personal and economic consequences, including impaired recovery, chronic pain after surgery, disability, and reduced quality of life; Indigenous peoples carry a disproportionate burden of chronic pain.[^1]

Physiology and Types of Pain

- **Nociception** involves four steps: transduction (conversion of noxious stimulus into an action potential), transmission (conduction to spinal cord and brain), perception (conscious awareness), and modulation (descending inhibition or facilitation).[^1]
- Peripheral sensitization and central sensitization (e.g., wind-up via NMDA receptors) amplify nociceptive input; preventing ongoing nociceptive input helps avoid chronic pain.[^1]
- **Nociceptive pain** arises from somatic or visceral tissue damage (e.g., surgical incision, arthritis, tumor); it is usually well responsive to non-opioids and opioids.[^1]
- **Neuropathic pain** results from nerve damage or abnormal processing (e.g., diabetic neuropathy, postherpetic neuralgia, phantom limb); it is often described as burning, shooting, or electric and typically requires adjuvant medications (antidepressants, anticonvulsants, etc.).[^1]
- **Acute pain** has sudden onset, is related to a specific injury/illness, diminishes with healing, and if unrelieved can contribute to chronic pain; **persistent (chronic) pain** lasts beyond expected healing (often >3 months), may lack clear tissue damage, and is closely linked to functional impairment, anxiety, and depression.[^1]

Pain Assessment

- Principles: pain is always subjective; self-report is the most reliable indicator; pain can exist without obvious physical cause; physiologic/behavioural signs are not specific and must not replace self-report.[^1]

- Pain should be assessed routinely in all settings, at least once daily, and more often for acute changes or following interventions.[^1]
- Initial and ongoing assessments use **OPQRSTUV** (Onset, Provocative/palliative, Quality, Region/radiation, Severity, Treatment/timing, Understanding, Values) and PAIN (Pattern, Area, Intensity, Nature).[^1]
- Tools include numeric rating scales, visual analogue scales, and faces scales; body diagrams help map location and radiation; terms other than "pain" (e.g., discomfort, soreness, ache) may be needed, especially with older adults.[^1]
- Assessment includes affective, behavioural, cognitive, and sociocultural components (impact on sleep, mood, function, relationships, beliefs about pain, coping strategies, meaning of pain).[^1]
- For people with cognitive impairment, behavioural indicators (vocalization, facial expression, body movements, tension, consolability) and input from caregivers are essential.[^1]

Pain Management – Principles and Pharmacology

- All patients have the right to effective pain management; nurses must assess, advocate, administer, monitor, document, and teach about pain management.[^1]
- A **multimodal, interprofessional** approach is recommended, combining pharmacologic and non-pharmacologic strategies, and involving patients/families in goal setting.[^1]
- **WHO analgesic ladder**: Step 1 – non-opioids (e.g., acetaminophen, NSAIDs) for mild pain; Step 2 – weak opioids plus non-opioids for mild–moderate pain; Step 3 – strong opioids for moderate–severe pain; steps are chosen based on pain severity, not necessarily strictly sequential.[^1]
- Non-opioids have a **ceiling effect**, do not cause physical dependence, and can be used alone or in combination; acetaminophen is preferred in many older adults but must stay within safe daily dose to avoid hepatotoxicity.[^1]
- NSAIDs block prostaglandin synthesis at different points in the arachidonic acid pathway; they are effective but carry risks (e.g., GI bleeding, renal dysfunction, cardiovascular events), especially in older adults.[^1]
- Step 3 opioids (e.g., morphine, hydromorphone, fentanyl, oxycodone) have no ceiling for analgesia but have dose-limiting adverse effects (constipation, nausea, sedation, respiratory depression, pruritus); constipation does not usually diminish with time and must be prevented.[^1]
- Equianalgesic dosing tables guide safe opioid rotation; methadone requires expert prescribing; meperidine is no longer recommended because of neurotoxicity (normeperidine) and lack of reversibility with naloxone.[^1]
- **Adjuvant analgesics** (e.g., antidepressants, anticonvulsants, local anesthetics, corticosteroids) target neuropathic pain, inflammation, or muscle spasm and support modulation of pain.[^1]

Ethical, Cultural, and Special Population Issues

- Ethical issues include fear of hastening death (addressed by the rule of double effect), condemnation of placebos for pain assessment/treatment, and the obligation to treat pain even at end of life.[^1]
- Older adults: start low and go slow; avoid or limit NSAIDs; watch for polypharmacy, cognitive effects, and falls; use non-pharmacological strategies and functional goals.[^1]
- Patients with **substance use disorders** still have the right to adequate pain control; higher doses may be needed due to tolerance, and mixed agonist-antagonists should be avoided because they can precipitate withdrawal.[^1]
- Indigenous peoples frequently encounter discrimination (e.g., assumptions of drug-seeking), contributing to undertreated pain; culturally safe, trauma-informed care is essential.[^1]

Chapter 19 – Fluid, Electrolyte, and Acid–Base Imbalance (Core Concepts – based on Unit 5.1 headings)

> Note: The Unit-5.1 extract emphasizes pain; only general headings for Chapter 19 appear, so these key points integrate standard Lewis med-surg F&E and acid–base content.

- **Body fluids** are distributed between intracellular and extracellular compartments; fluid balance is regulated by thirst, hormones (ADH, aldosterone, natriuretic peptides), and kidney function.[^2]
- **Isotonic, hypotonic, and hypertonic solutions** differ in osmolality and shift water between ECF and ICF; nurses must choose and monitor IV fluids based on patient status (e.g., isotonic for volume depletion, hypertonic for severe hyponatremia, hypotonic cautiously when cells are dehydrated).[^2]
- Common **electrolyte imbalances** (Na, K, Ca, Mg, phosphate) have characteristic causes (e.g., GI loss, renal failure, diuretics) and manifestations (e.g., arrhythmias, neuromuscular changes, mental-status changes) that guide assessment and interventions.[^2]
- **Acid–base balance** is reflected by arterial blood gases; respiratory imbalances result from CO_2 changes, metabolic imbalances from HCO_3^-; compensation and mixed disorders must be recognized to guide treatment.[^2]
- Nursing care includes close monitoring of intake and output, daily weights, vital signs, lab values, IV therapy, and focused assessments (e.g., lung sounds, edema, mental status) plus patient teaching about diet, fluid restriction, and medications.[^2]

Chapter 20 – Nursing Management: Pre-operative Care

> Note: Unit-5.1 references this chapter but the detailed text is not in the extract; key concepts follow standard Lewis pre-op content.

- Pre-operative nursing care aims to **establish a baseline**, identify and manage risks, and prepare patients physically and psychologically for surgery.[^2]
- Assessment includes comprehensive health history (medical, surgical, medication, allergy, substance use, coping, support systems), focused physical exam, and review of diagnostic results.[^2]
- Special attention is given to **cardiovascular, respiratory, renal, hepatic, endocrine, and immune** function, as well as nutritional status, to anticipate peri-operative complications.[^2]
- Nurses verify **informed consent**, ensure pre-operative teaching (what to expect, pain control, breathing and leg exercises, equipment like drains or catheters), and reduce anxiety through clear communication and support.[^2]
- Pre-op preparation often includes NPO status, bowel prep when indicated, skin antisepsis, removal of prostheses and jewelry, and administration of ordered pre-op medications (e.g., antibiotics, sedatives) with safety checks.[^2]
- Immediately before surgery, nurses complete checklists, confirm identification, allergies, site/side (time-out), and accompany or transfer the patient safely to the OR, handing off accurate information to the peri-operative team.[^2]

Chapter 22 – Nursing Management: Post-operative Care

> Note: Detailed chapter text is not shown in the Unit-5.1 extract; key points summarize standard Lewis post-op content.

- Post-operative care begins in the PACU and continues on the surgical unit, focusing first on **airway, breathing, and circulation** (ABCs), neurologic status, and pain control.[^2]
- Initial priorities include maintaining a patent airway, monitoring respiratory rate, depth, oxygen saturation, and auscultating lungs; early detection of hypoventilation, hypoxemia, or airway obstruction is critical.[^2]
- Hemodynamic monitoring (vital signs, skin colour and temperature, urine output, IV fluids) detects hypovolemia, bleeding, and shock; surgical drains and dressings are checked for amount and type of drainage.[^2]
- Common early post-op complications include atelectasis, pneumonia, DVT/PE, hemorrhage, shock, urinary retention, ileus, nausea/vomiting, and pain; later complications include infection and dehiscence/evisceration.[^2]
- Nursing interventions emphasize **early mobilization**, deep breathing and coughing (or incentive spirometry as ordered), leg exercises, prophylaxis for thrombosis, adequate pain management, and progressive diet and activity as tolerated.[^2]

- Wound care includes proper dressing changes, monitoring for signs of infection (redness, warmth, swelling, purulent drainage, fever), and supporting patient teaching on incision care and activity restrictions after discharge.[^2]
- Discharge teaching addresses medications, activity and lifting limits, diet, signs and symptoms that require urgent medical review, follow-up appointments, and who to contact with concerns.[^2]

References

1. Unit-5.1.pdf - Pain
2. Lewis's Medical-Surgical Nursing in Canada, 5th Edition - Updated Substance Use chapter covers stigmatizing behaviours by health care ... 11 Substance Use 12 ...